

LIMITS

So very roughly speaking, “Differential Calculus” is the study of how a function changes as its input changes. The mathematical object we use to describe this is the “derivative” of a function. To properly describe what this thing is we need some machinery; in particular we need to define what we mean by “tangent” and “limit”. We’ll get back to defining the derivative in Chapter 2.

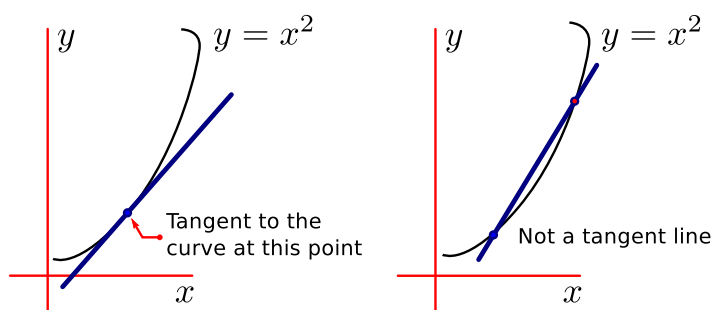
1.1 ▲ Drawing Tangents and a First Limit

Our motivation for developing “limit” — being the title and subject of this chapter — is going to be two related problems of drawing tangent lines and computing velocity.

Now — our treatment of limits is not going to be completely mathematically rigorous, so we won’t have too many formal definitions. There will be a few mathematically precise definitions and theorems as we go, but we’ll make sure there is plenty of explanation around them.

Let us start with the “tangent line” problem. Of course, we need to define “tangent”, but we won’t do this formally. Instead let us draw some pictures.

Figure 1.1.1.

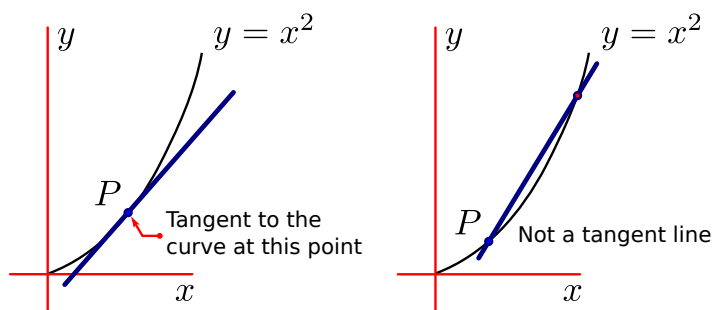


Here we have drawn two very rough sketches of the curve $y = x^2$ for $x \geq 0$. These are not very good sketches for a couple of reasons

- The curve in the figure does not pass through $(0,0)$, even though $(0,0)$ lies on $y = x^2$.
- The top-right end of the curve doubles back on itself and so fails the vertical line test that all functions must satisfy¹ — for each x -value there is exactly one y -value for which (x, y) lies on the curve $y = x^2$.

So let's draw those more carefully.

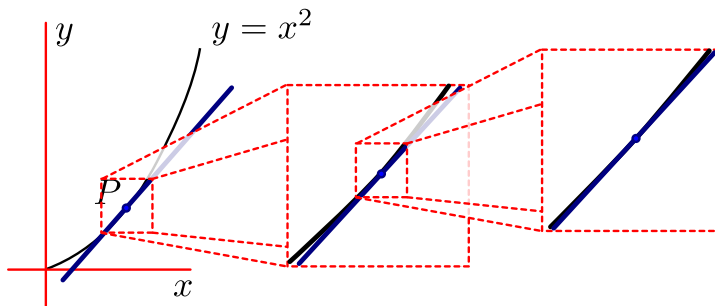
Figure 1.1.2.



Sketches of the curve $y = x^2$. (left) shows a tangent line, while (right) shows a line that is not a tangent.

These are better. In both cases we have drawn $y = x^2$ (carefully) and then picked a point on the curve — call it P . Let us zoom in on the “good” example:

Figure 1.1.3.



We see that, the more we zoom in on the point P , the more the graph of the function (drawn in black) looks like a straight line — that line is the tangent line (drawn in blue).

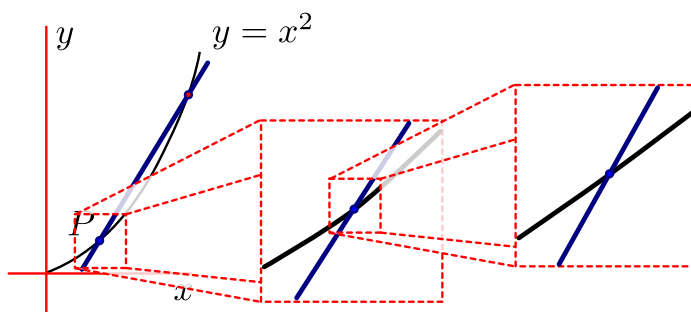
We see that as we zoom in on the point P , the graph of the function looks more and more like a straight line. If we kept on zooming in on P then the graph of the function would be indistinguishable from a straight line. That line is the tangent line (which we

¹ Take a moment to go back and reread Definition [0.4.1](#).

have drawn in blue). A little more precisely, the blue line is “the tangent line to the function at P ”. We have to be a little careful, because if we zoom in at a different point, then we will find a different tangent line.

Now let’s zoom in on the “bad” example we see that the blue line looks very different from the function; because of this, the blue line is not the tangent line at P .

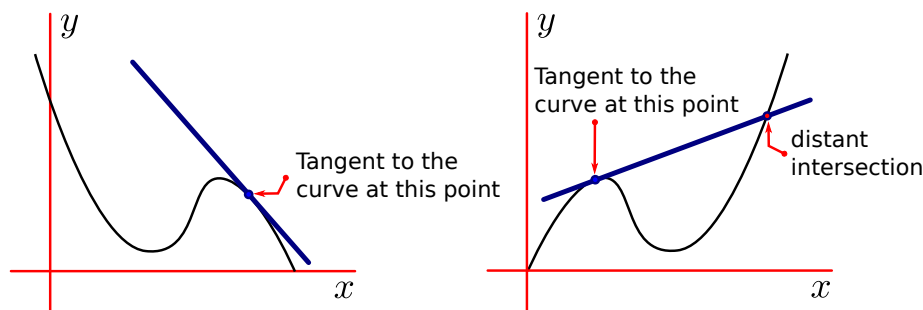
Figure 1.1.4.



Zooming in on P we see that the function (drawn in black) looks more and more like a straight line — however it is not the same line as that drawn in blue. Because of this the blue line is not the tangent line.

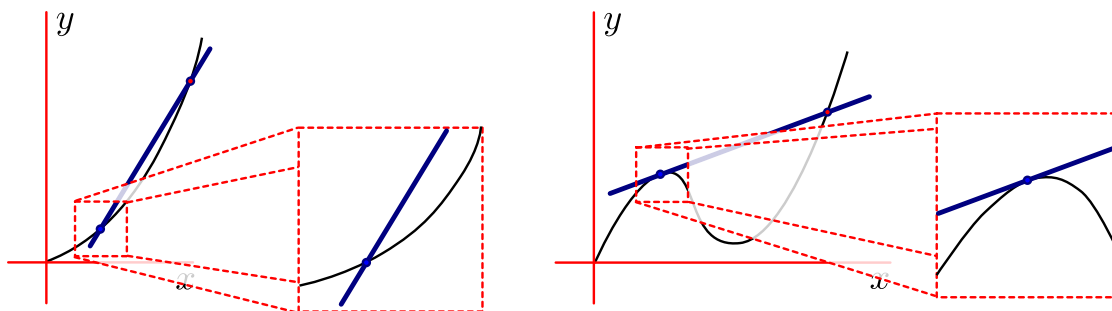
Here are a couple more examples of tangent lines

Figure 1.1.5.



More examples of tangent lines.

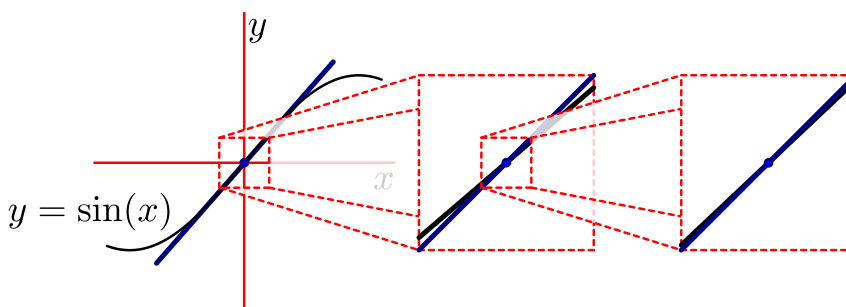
The one on the left is very similar to the good example on $y = x^2$ that we saw above, while the one on the right is different — it looks a little like the “bad” example, in that it crosses our function the curve at some distant point. Why is the line in Figure 1.1.5(right) a tangent while the line in Figure 1.1.2(right) not a tangent? To see why, we should again zoom in close to the point where we are trying to draw the tangent.

Figure 1.1.6.

As we saw above in Figure 1.1.4, when we zoom in around our example of “not a tangent line” we see that the straight line looks very different from the curve at the “point of tangency” — i.e. where we are trying to draw the tangent. The line drawn in Figure 1.1.5(right) looks more and more like the function as we zoom in.

This example raises an important point — when we are trying to draw a tangent line, we don’t care what the function does a long way from the point; the tangent line to the curve at a particular point P , depends only on what the function looks like close to that point P .

To illustrate this consider the sketch of the function $y = \sin(x)$ and its tangent line at $(x, y) = (0, 0)$:

Figure 1.1.7.

As we zoom in, the graph of $\sin(x)$ looks more and more like a straight line — in fact it looks more and more like the line $y = x$. We have also sketched this tangent line. What makes this example a little odd is that the tangent line crosses the function. In the examples above, our tangent lines just “kissed” the curve and did not cross it (or at least did not cross it nearby).

Using this idea of zooming in at a particular point, drawing a tangent line is not too hard. However, finding the equation of the tangent line presents us with a few challenges. Rather than leaping into the general theory, let us do a specific example. Let us find

the equation of the tangent line to the curve $y = x^2$ at the point P with coordinates² $(x, y) = (1, 1)$.

To find the equation of a line we either need

- the slope of the line and a point on the line, or
- two points on the line, from which we can compute the slope via the formula

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

and then write down the equation for the line via a formula such as

$$y = m \cdot (x - x_1) + y_1.$$

We cannot use the first method because we do not know what the slope of the tangent line should be. To work out the slope we need calculus — so we'll be able to use this method once we get to the next chapter on "differentiation".

It is not immediately obvious how we can use the second method, since we only have one point on the curve, namely $(1, 1)$. However we can use it to "sneak up" on the answer. Let's approximate the tangent line, by drawing a line that passes through $(1, 1)$ and some nearby point — call it Q . Here is our recipe:

- We are given the point $P = (1, 1)$ and we are told

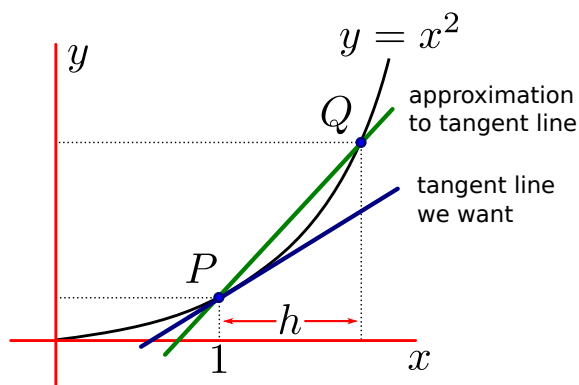
Find the tangent line to the curve $y = x^2$ that passes through $P = (1, 1)$.

- We don't quite know how to find a line given just 1 point, however we do know how to find a line passing through 2 points. So pick another point on the curves whose coordinates are very close to P . Now rather than picking some actual numbers, I am going to write our second point as $Q = (1 + h, (1 + h)^2)$. That is, a point Q whose x -coordinate is equal to that of P plus a little bit — where the little bit is some small number h . And since this point lies on the curve $y = x^2$, and Q 's x -coordinate is $1 + h$, Q 's y -coordinate must be $(1 + h)^2$.

If having h as a variable rather than a number bothers you, start by thinking of h as 0.1.

- A picture of the situation will help.

2 Note that the *coordinates* (x, y) is an ordered pair of two numbers x and y . Traditionally the first number is called the *abscissa* while the second is the *ordinate*, but these terms are a little archaic. It is now much more common to hear people refer to the first number as the *x-coordinate* and the second as *y-coordinate*.

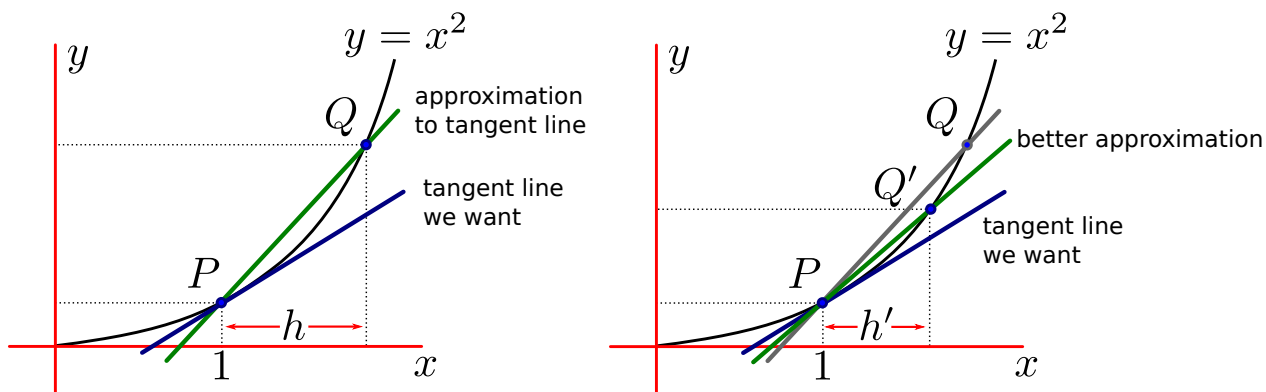
Figure 1.1.8.

- This line that passes through the curve in two places P and Q is called a “secant line”.
- The slope of the line is then

$$\begin{aligned}
 m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{(1+h)^2 - 1}{(1+h) - 1} = \frac{1 + 2h + h^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h
 \end{aligned}$$

where we have expanded $(1+h)^2 = 1 + 2h + h^2$ and then cleaned up a bit.

Now this isn't our tangent line because it passes through 2 nearby points on the curve — however it is a reasonable approximation of it. Now we can make that approximation better and so “sneak up” on the tangent line by considering what happens when we move this point Q closer and closer to P . i.e. make the number h closer and closer to zero.

Figure 1.1.9.

First look at the picture. The original choice of Q is on the left, while on the right we have drawn what happens if we choose h' to be some number a little smaller than h , so

that our point Q becomes a new point Q' that is a little closer to P . The new approximation is better than the first.

So as we make h smaller and smaller, we bring Q closer and closer to P , and make our secant line a better and better approximation of the tangent line. We can observe what happens to the slope of the line as we make h smaller by plugging some numbers into our formula $m = 2 + h$:

$$\begin{array}{ll} h = 0.1 & m = 2.1 \\ h = 0.01 & m = 2.01 \\ h = 0.001 & m = 2.001. \end{array}$$

So again we see that as this difference in x becomes smaller and smaller, the slope appears to be getting closer and closer to 2. We can write this more mathematically as

$$\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} = 2$$

This is read as

The limit, as h approaches 0, of $\frac{(1+h)^2 - 1}{h}$ is 2.

This is our first limit! Notice that we can see this a little more clearly with a quick bit of algebra:

$$\begin{aligned} \frac{(1+h)^2 - 1}{h} &= \frac{(1 + 2h + h^2) - 1}{h} \\ &= \frac{2h + h^2}{h} \\ &= 2 + h \end{aligned}$$

So it is not unreasonable to expect that

$$\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} = \lim_{h \rightarrow 0} (2 + h) = 2.$$

Our tangent line can be thought of as the end of this process — namely as we bring Q closer and closer to P , the slope of the secant line comes closer and closer to that of the tangent line we want. Since we have worked out what the slope is — that is the limit we saw just above — we now know the slope of the tangent line is 2. Given this, we can work out the equation for the tangent line.

- The equation for the line is $y = mx + c$. We have 2 unknowns m and c — so we need 2 pieces of information to find them.
- Since the line is tangent to $P = (1, 1)$ we know the line must pass through $(1, 1)$. From the limit we computed above, we also know that the line has slope 2.
- Since the slope is 2 we know that $m = 2$. Thus the equation of the line is $y = 2x + c$.
- We know that the line passes through $(1, 1)$, so that $y = 2x + c$ must be 1 when $x = 1$. So $1 = 2 \cdot 1 + c$, which forces $c = -1$.

So our tangent line is $y = 2x - 1$.

1.2 ▲ Another Limit and Computing Velocity

Computing tangent lines is all very well, but what does this have to do with applications or the “Real World”? Well - at least initially our use of limits (and indeed of calculus) is going to be a little removed from real world applications. However as we go further and learn more about limits and derivatives we will be able to get closer to real problems and their solutions.

So stepping just a little closer to the real world, consider the following problem. You drop a ball from the top of a very very tall building. Let t be elapsed time measured in seconds, and $s(t)$ be the distance the ball has fallen in metres. So $s(0) = 0$.

Quick aside: there is quite a bit going on in the statement of this problem. We have described the general picture — tall building, ball, falling — but we have also introduced notation, variables and units. These will be common first steps in applications and are necessary in order to translate a real world problem into mathematics in a clear and consistent way.

Galileo³ worked out that $s(t)$ is a quadratic function:

$$s(t) = 4.9t^2.$$

The question that is posed is

How fast is the ball falling after 1 second?

Now before we get to answering this question, we should first be a little more precise. The wording of this question is pretty sloppy for a couple of reasons:

- What we do mean by “after 1 second”? We know the ball will move faster and faster as time passes, so after 1 second it does not fall at one fixed speed.
- As it stands a reasonable answer to the question would be just “really fast”. If the person asking the question wants a numerical answer it would be better to ask “At what speed” or “With what velocity”.

We should also be careful using the words “speed” and “velocity” — they are not interchangeable.

- Speed means the distance travelled per unit time and is always a non-negative number. An unmoving object has speed 0, while a moving object has positive speed.
- Velocity, on the other hand, also specifies the direction of motion. In this text we will almost exclusively deal with objects moving along straight lines. Because of this

3 Perhaps one of the most famous experiments in all of physics is Galileo’s leaning tower of Pisa experiment, in which he dropped two balls of different masses from the top of the tower and observed that the time taken to reach the ground was independent of their mass. This disproved Aristotle’s assertion that heavier objects fall faster. It is quite likely that Galileo did not actually perform this experiment. Rather it was a thought-experiment. However a quick glance at Wikipedia will turn up some wonderful footage from the Apollo 15 mission showing a hammer and feather being dropped from equal height hitting the moon’s surface at the same time. Finally, Galileo determined that the speed of falling objects increases at a constant rate, which is equivalent to the formula stated here, but it is unlikely that he wrote down an equation exactly as it is here.

velocities will be positive or negative numbers indicating which direction the object is moving along the line. We will be more precise about this later⁴.

A better question is

What is the velocity of the ball precisely 1 second after it is dropped?

or even better:

What is the velocity of the ball at the 1 second mark?

This makes it very clear that we want to know what is happening at exactly 1 second after the ball is dropped.

There is something a little subtle going on in this question. In particular, what do we mean by the velocity at $t = 1$? Surely if we freeze time at $t = 1$ second, then the object is not moving at all? This is definitely *not* what we mean.

If an object is moving at a constant velocity⁵ in the positive direction, then that velocity is just the distance travelled divided by the time taken. That is

$$v = \frac{\text{distance moved}}{\text{time taken}}$$

An object moving at constant velocity that moves 27 metres in 3 seconds has velocity

$$v = \frac{27m}{3s} = 9m/s.$$

When velocity is constant everything is easy.

However, in our falling object example, the object is being acted on by gravity and its speed is definitely not constant. Instead of asking for *THE* velocity, let us examine the “average velocity” of the object over a certain window of time. In this case the formula is very similar

$$\text{average velocity} = \frac{\text{distance moved}}{\text{time taken}}$$

But now I want to be more precise, instead write

$$\text{average velocity} = \frac{\text{difference in distance}}{\text{difference in time}}$$

Now in spoken English we haven’t really changed much — the distance moved is the difference in position, and the time taken is just the difference in time — but the latter is more mathematically precise, and is easy to translate into the following equation

$$\text{average velocity} = \frac{s(t_2) - s(t_1)}{t_2 - t_1}.$$

4 Getting the sign of velocity wrong is a very common error — you should be careful with it.

5 Newton’s first law of motion states that an object in motion moves with constant velocity unless a force acts on it — for example gravity or friction.

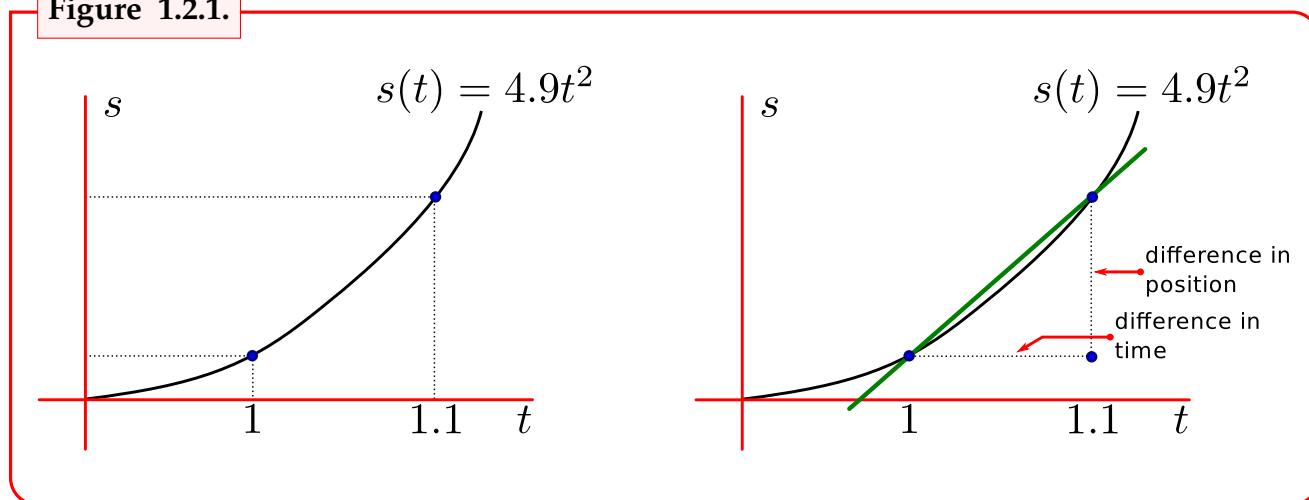
This is the formula for the average velocity of our object between time t_1 and t_2 . The denominator is just the difference between these times and the numerator is the difference in position — i.e. position at time t_1 is just $s(t_1)$ and position at time t_2 is just $s(t_2)$.

So what is the average velocity of the falling ball between 1 and 1.1 seconds? All we need to do now is plug some numbers into our formula

$$\begin{aligned}\text{average velocity} &= \frac{\text{difference in position}}{\text{difference in time}} \\ &= \frac{s(1.1) - s(1)}{1.1 - 1} \\ &= \frac{4.9(1.1)^2 - 4.9(1)}{0.1} = \frac{4.9 \times 0.21}{0.1} = 10.29 \text{ m/s}\end{aligned}$$

And we have our average velocity. However there is something we should notice about this formula and it is easier to see if we sketch a graph of the function $s(t)$

Figure 1.2.1.



So on the left I have drawn the graph and noted the times $t = 1$ and $t = 1.1$. The corresponding positions on the axes and the two points on the curve. On the right I have added a few more details. In particular I have noted the differences in position and time, and the line joining the two points. Notice that the slope of this line is

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{difference in } s}{\text{difference in } t}$$

which is precisely our expression for the average velocity.

Let us examine what happens to the average velocity as we look over smaller and smaller time-windows.

time window	average velocity
$1 \leq t \leq 1.1$	10.29
$1 \leq t \leq 1.01$	9.849
$1 \leq t \leq 1.001$	9.8049
$1 \leq t \leq 1.0001$	9.80049

LIMITS

1.1▲ Drawing Tangents and a First Limit

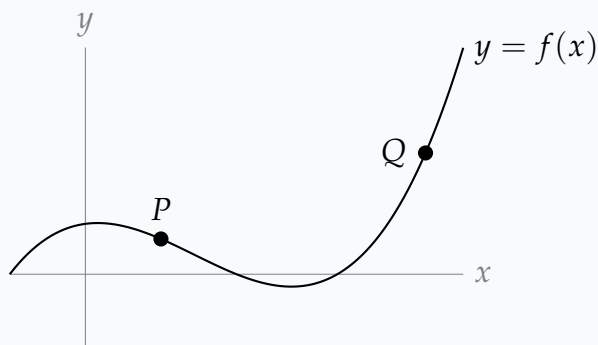
Exercises

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►► Stage 1

Q[1]: On the graph below, draw:

- (a) The tangent line to $y = f(x)$ at P ,
- (b) the tangent line to $y = f(x)$ at Q , and
- (c) the secant line to $y = f(x)$ through P and Q .



Q[2]: Suppose a curve $y = f(x)$ has tangent line $y = 2x + 3$ at the point $x = 2$.

- (a) True or False: $f(2) = 7$
- (b) True or False: $f(3) = 9$

Q[3]: Let L be the tangent line to a curve $y = f(x)$ at some point P . How many times will L intersect the curve $y = f(x)$?

1.2▲ Another Limit and Computing Velocity

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: As they are used in this section, what is the difference between speed and velocity?

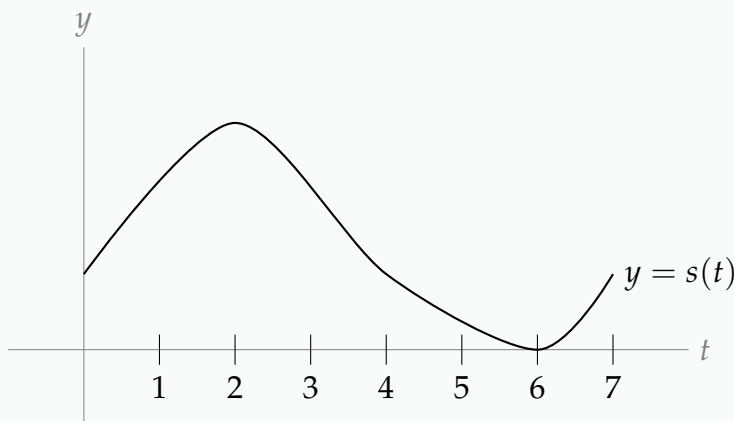
Q[2]: Speed can never be negative; can it be zero?

Q[3]: Suppose you wake up in the morning in your room, then you walk two kilometres to school, walk another two kilometres to lunch, walk four kilometres to a coffee shop to study, then return to your room until the next morning. In the 24 hours from morning to morning, what was your average velocity? (In CLP-1, we are considering functions of one variable. So, at this stage, think of our whole world as being contained in the x -axis.)

Q[4]: Suppose you drop an object, and it falls for a few seconds. Which is larger: its speed at the one second mark, or its average speed from the zero second mark to the one second mark?

Q[5]: The position of an object at time t is given by $s(t)$. Then its average velocity over the time interval $t = a$ to $t = b$ is given by $\frac{s(b) - s(a)}{b - a}$. Explain why this fraction also gives the slope of the secant line of the curve $y = s(t)$ from the point $t = a$ to the point $t = b$.

Q[6]: Below is the graph of the position of an object at time t . For what periods of time is the object's velocity positive?



►► Stage 2

Q[7]: Suppose the position of a body at time t (measured in seconds) is given by $s(t) = 3t^2 + 5$.

- (a) What is the average velocity of the object from 3 seconds to 5 seconds?
- (b) What is the velocity of the object at time $t = 1$?

Q[8]: Suppose the position of a body at time t (measured in seconds) is given by $s(t) = \sqrt{t}$.

- (a) What is the average velocity of the object from $t = 1$ second to $t = 9$ seconds?
- (b) What is the velocity of the object at time $t = 1$?
- (c) What is the velocity of the object at time $t = 9$?

1.3▲ The Limit of a Function

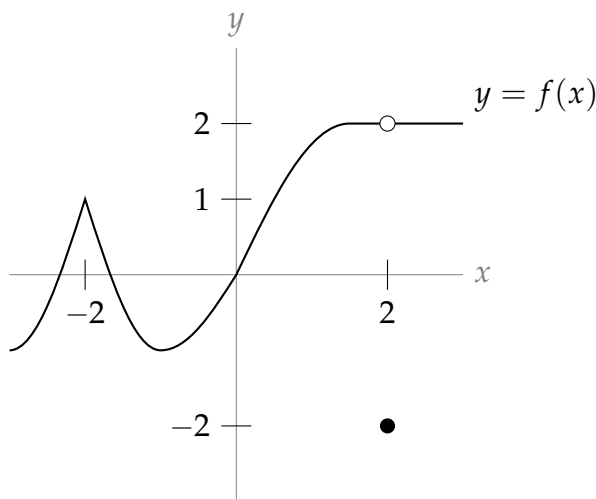
Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: Given the function shown below, evaluate the following:

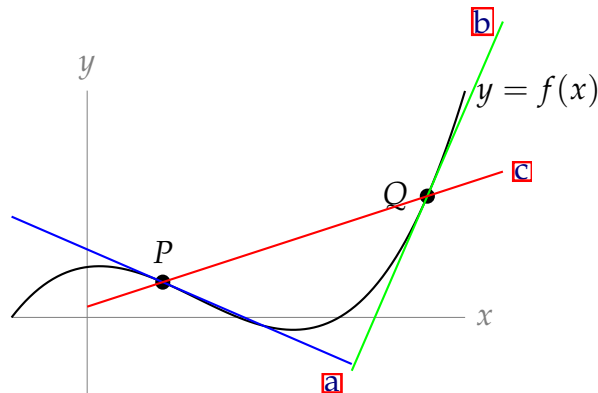
- (a) $\lim_{x \rightarrow -2} f(x)$
- (b) $\lim_{x \rightarrow 0} f(x)$
- (c) $\lim_{x \rightarrow 2} f(x)$



Q[2]: Given the function shown below, evaluate $\lim_{x \rightarrow 0} f(x)$.

Solutions to Exercises 1.1 — Jump to [TABLE OF CONTENTS](#)

S-1:

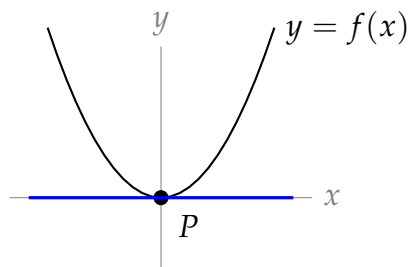


The tangent line to $y = f(x)$ at a point should go through the point, and be “in the same direction” as f at that point. The secant line through P and Q is simply the straight line passing through P and Q .

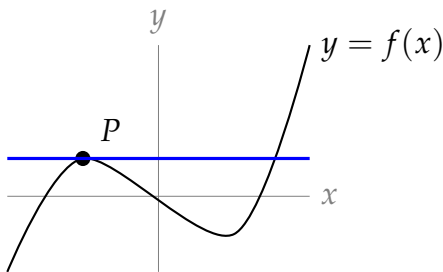
S-2:

- (a) True: since $y = 2x + 3$ is the tangent line to $y = f(x)$ at the point $x = 2$, this means the function and the tangent line have the same value at $x = 2$. So $f(2) = 2(2) + 3 = 7$.
- (b) In general, this is false. We are only guaranteed that the curve $y = f(x)$ and its tangent line $y = 2x + 3$ agree at $x = 2$. The functions $f(x)$ and $2x + 3$ may or may not take the same values when $x \neq 2$. For example, if $f(x) = 2x + 3$, then of course $f(x)$ and $2x + 3$ agree for all values of x . But if $f(x) = 2x + 3 + (x - 2)^2$, then $f(x)$ and $2x + 3$ agree only for $x = 2$.

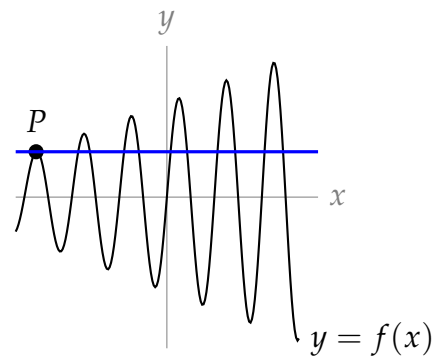
S-3: Since the tangent line to the curve at point P passes through point P , the curve and the tangent line touch at point P . So, they must intersect at least once. By drawing various examples, we can see that different curves may touch their tangent lines exactly once, exactly twice, exactly three times, etc.



Blue tangent line touches curve only once, at P



Blue tangent line touches curve twice



Blue tangent line touches curve many times

Solutions to Exercises [1.2](#) — Jump to [TABLE OF CONTENTS](#)

S-1: Speed is nonnegative; velocity has a sign (positive or negative) that indicates direction.

S-2: Yes—an object that is not moving has speed 0.

S-3: Since you started and ended in the same place, your difference in position was 0, and your difference in time was 24 hours. So, your average velocity was $\frac{0}{24} = 0$ kph.

S-4: Objects accelerate as they fall — their speed gets bigger and bigger. So in the entire first second of falling, the object is at its fastest at the one-second mark. We have defined the average speed over a time interval to be $\frac{\text{distance moved}}{\text{time taken}}$. We do not yet know how to compute the distance travelled by an object whose speed is not constant. But our intuition certainly says that the distance travelled is no more than the maximum speed times the time taken, so that the average speed is no larger than the maximum speed. In the next CLP-1 chapter we will verify that intuition mathematically. See Example [2.13.6](#) in the CLP-1 text. So the average speed will be a smaller number than the speed at the one-second mark, which is the maximum speed.

S-5: The slope of a curve is given by $\frac{\text{change in vertical component}}{\text{change in horizontal component}}$. The change in the vertical component is exactly $s(b) - s(a)$, and the change in the horizontal component is exactly $b - a$.

S-6: The velocity is positive when the object is going in the increasing direction; it is going “up” on the graph when t is between 0 and 2, and when t is between 6 and 7. So, the velocity is positive when t is in $(0, 2) \cup (6, 7)$.

S-7:

(a) Average velocity: $\frac{\text{change in position}}{\text{change in time}} = \frac{s(5) - s(3)}{5 - 3} = \frac{(3 \cdot 5^2 + 5) - (3 \cdot 3^2 + 5)}{5 - 3} = 24$ units per second.

(b) From the notes, we know the velocity of an object at time a is

$$v(a) = \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

So, in our case:

$$v(1) = \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = \lim_{h \rightarrow 0} \frac{[3(1+h)^2 + 5] - [3(1)^2 + 5]}{h} = \lim_{h \rightarrow 0} \frac{6h + 3h^2}{h} = \lim_{h \rightarrow 0} 6 + 3h = 6$$

So the velocity when $t = 1$ is 6 units per second.

S-8:

(a) Average velocity: $\frac{\text{change in position}}{\text{change in time}} = \frac{s(9) - s(1)}{9 - 1} = \frac{3 - 1}{9 - 1} = \frac{1}{4}$ units per second.

(b) From the notes, we know the velocity of an object at time a is

$$v(a) = \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

So, in our case:

$$\begin{aligned} v(1) &= \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{1+h} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{1+h} - 1}{h} \cdot \left(\frac{\sqrt{1+h} + 1}{\sqrt{1+h} + 1} \right) \\ &= \lim_{h \rightarrow 0} \frac{(1+h) - 1}{h(\sqrt{1+h} + 1)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{1+h} + 1} = \frac{1}{2} \text{ units per second} \end{aligned}$$

(c)

$$\begin{aligned} v(9) &= \lim_{h \rightarrow 0} \frac{s(9+h) - s(9)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} \cdot \left(\frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3} \right) \\ &= \lim_{h \rightarrow 0} \frac{(9+h) - 9}{h(\sqrt{9+h} + 3)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{9+h} + 3} = \frac{1}{6} \text{ units per second} \end{aligned}$$

Remark: the average velocity is *not* the average of the two instantaneous velocities.

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S-1:

- (a) $\lim_{x \rightarrow -2} f(x) = 1$: as x gets very close to -2 , y gets very close to 1.
- (b) $\lim_{x \rightarrow 0} f(x) = 0$: as x gets very close to 0, y also gets very close to 0.
- (c) $\lim_{x \rightarrow 2} f(x) = 2$: as x gets very close to 2, y gets very close to 2. We ignore the value of the function where x is exactly 2.

S-2: The limit does not exist. As x approaches 0 *from the left*, y approaches -1 ; as x approaches 0 *from the right*, y approaches 1. This tells us $\lim_{x \rightarrow 0^-} f(x) = -1$ and $\lim_{x \rightarrow 0^+} f(x) = 1$, but neither of these are what the question asked. Since the limits from left and right do not agree, the limit does not exist. Put another way, there is no single number y approaches as x approaches 0, so the limit $\lim_{x \rightarrow 0} f(x)$ does not exist.

S-3:

- (a) $\lim_{x \rightarrow -1^-} f(x) = 2$: as x approaches -1 from the left, y approaches 2. It doesn't matter that the function isn't defined at $x = -1$, and it doesn't matter what happens to the right of $x = -1$.
- (b) $\lim_{x \rightarrow -1^+} f(x) = -2$: as x approaches -1 from the right, y approaches -2 . It doesn't matter that the function isn't defined at -1 , and it doesn't matter what happens to the left of -1 .
- (c) $\lim_{x \rightarrow -1} f(x) = \text{DNE}$: since the limits from the left and right don't agree, the limit does not exist.
- (d) $\lim_{x \rightarrow -2^+} f(x) = 0$: as x approaches -2 from the right, y approaches 0. It doesn't matter that the function isn't defined at 2, or to the left of 2.
- (e) $\lim_{x \rightarrow 2^-} f(x) = 0$: as x approaches 2 from the left, y approaches 0. It doesn't matter that the function isn't defined at 2, or to the right of 2.

S-4: Many answers are possible; here is one.